

Leonardo Marciaga

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SUMMARY

Applied mathematics student with experience in data analysis and Large Language Models. Highly adaptable and strong at technical problem-solving, with passion for effective communication of technical concepts.

EDUCATION

Illinois Institute of Technology, Chicago, IL August 2023 – May 2027 (expected)
Bachelor of Science, Major in Applied Mathematics, Minor in Artificial Intelligence
Cumulative GPA: **4.0/4.0**

SKILLS

Programming: Python | R | HTML | CSS
Data manipulation and analysis: pandas | scikit-learn | tidyverse
Data visualization: matplotlib | ggplot2

RELEVANT EXPERIENCE

Research Intern June – August 2025
Institute for Pure and Applied Mathematics (IPAM), UCLA, Los Angeles, CA

- Worked with the RAND Corporation using Large Language Models to augment survey data for epidemic modeling.
- Developed automated synthetic data generation pipeline in Python employing data analysis results and LLM APIs to efficiently produce augmented datasets achieving 80% similarity to real-world overall statistical trends.
- Identified underlying cluster patterns in complex longitudinal survey data encompassing nearly 10 years and 2000+ respondents, applying algorithms such as K-medoids, hierarchical and spectral clustering using R.
- Communicated findings on dataset structure to stakeholders using visualization techniques such as t-SNE and UMAP.

ADDITIONAL EXPERIENCE

Research Assistant August 2024 – Present
Department of Applied Mathematics, Illinois Tech, Chicago, IL
Topic: List coloring number of Cartesian products of graphs with complete bipartite factors

- Explored methods from literature in discrete mathematics and graph theory to help improve advisors' previous work.
- Co-authored technical publication on project results. <https://arxiv.org/abs/2509.16733>

Associate Instructor January 2021 – Present
Panamanian Mathematical Olympiad Foundation, Panama

- Led training sessions for Panama's math contest training program, leading to award-winning students at contests such as the Ibero-American Mathematical Olympiad and the Pan-American Girls' Mathematical Olympiad.
- Mentored students, organized training schedules and created class material in a variety of mathematical areas.

Teaching Assistant August 2024 – December 2024
Department of Computer Science, Illinois Tech, Chicago, IL

- Taught basic programming concepts to 15 non-CS students, providing personalized assistance and feedback.

OTHER HONORS AND ACHIEVEMENTS

1st Place Team Award, Oral Presentation January 2023
2022 Samsung Innovation Campus Hackathon, Panama

- Developed an image classifier application for detecting diseases in 9 important agricultural products in the region.
- Awarded first place among more than 30 participating Panamanian teams.

Mathematical Competitions:

- William Lowell Putnam Mathematical Competition (Putnam): Top 500 finisher (2024)
- International Mathematical Olympiad (IMO): Honorable mention (2018, 2019, 2020)
- Asian-Pacific Mathematical Olympiad (APMO): Bronze medal (2020), honorable mention (2018, 2019)
- Ibero-American Mathematical Olympiad: Silver medal (2019)